25900232 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBCH202 - TECHNIQUES IN BIOCHEMISTRY AND FORENSIC SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The purpose of hybridization in Northern blotting is [U] / [CO1]

   a). To degrade excess RNA b). To label the RNA

   c). To bind a complementary probe to the RNA of interest d). To replicate the RNA

2. The separation in gel filtration chromatography is based on_____. [U] / [CO1]

   a). Charge b). Size

   c). Affinity d). Partition

3. Dark field microscopy is used in [U] / [CO2]

   a). Polarizing microscope b). Metallurgical microscope

   c). Simple microscope d). None of these

4. Choose the person who coined the term "cell". [K] / [CO2]

a). Robert Hooke b). Louis Pasteur

c). Anton van Leeuwenhoek d). Robert Koch

5. The type of DNA found in hair shaft is [U] / [CO3]

a). Nuclear DNA b). Mitochondrial DNA

c). Both A and B d). None of these

[1x5 = 5]

Part B

Fill In The Blank Questions

Answer all questions

Each question carries **2** marks

6. SDS is used as_____ in polyacrylamide gel electrophoresis. [K] / [CO1]

7. In affinity chromatography, the target molecule binds specifically to the_____. [K] / [CO1]

8. _____is the part of a centrifuge that contains tube holders and drive shaft. [U] / [CO2]

9. The first electron microscope was constructed by_____. [K] / [CO2]

10. In maternity testing, the child's DNA is compared with that of the_____ mother. [U] / [CO3]

[2x5 = 10]

Part C

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Explain why two dimensional paper chromatography is more useful than one dimensional chromatography in separating amino acids. [U] / [CO1]

12. Discuss the principle of Southern blotting. [U] / [CO1]

13. Explain the advantages of spectrophotometer over colorimeter. [U] / [CO2]

14. List any three major types of spectroscopy. [K] / [CO2]

15. Differentiate between direct ELISA and sandwich ELISA. [U] / [CO3]

16. Discuss the advantages and limitations of RIA. [U] / [CO3]

17. Recognize the function of the primer in PCR and give the typical temperature for the annealing step in PCR. [K] / [CO3]

[3x5 = 15]

Part D

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18. Define Western blotting and recognize its significance in protein analysis. [K] / [CO1]

19. Describe the procedure and applications of TLC. [U] / [CO1]

20. Illustrate the significance of ultracentrifuges in biochemistry and molecular biology studies. [A] / [CO2]

21. Explain Beer Lambert's Law and identify the various components present in a colorimeter. [U] / [CO2]

22. Discuss the various steps involved in DNA fingerprinting for maternity testing. [U] / [CO3]

23. Discuss how skin cells, urine, faeces and teeth became valuable sources of DNA in forensic investigations. [U] / [CO3]

[5x4 = 20]